

Commissioned Review: Bacterial Putrescine Biosynthesis by Arginine and Ornithine Routes in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK, NCBI taxon 160488, proteome UP000000556) **Target bucket:** KEGG `ppu00330` — "Arginine and proline metabolism" (30 primary genes; 39 candidates) **Module under review:** Putrescine supply through two alternative amino-acid routes — arginine route (SpeA → agmatine → SpeB → putrescine) and direct ornithine route (SpeC → putrescine). Spermidine synthesis, putrescine uptake, and putrescine catabolism are out of scope. **Date:** 2026-08-13 · Supersedes an Iteration-1 error (aguB = PP_1639), corrected below.

1. Executive Summary

The putrescine-biosynthesis module is **satisfiable** in *Pseudomonas putida* KT2440. The genome encodes high-confidence orthologs for every core step of both canonical routes. In the **arginine route**, biosynthetic arginine decarboxylase SpeA (PP_0567, K01585, EC 4.1.1.19) converts L-arginine to agmatine, and agmatinase SpeB (PP_2196, K01480, EC 3.5.3.11) hydrolyzes agmatine directly to putrescine + urea. In the **direct ornithine route**, ornithine decarboxylase SpeC (PP_0864, K01581, EC 4.1.1.17) decarboxylates L-ornithine to putrescine in a single step. Module steps 1 (SpeA agmatine formation), 2 (SpeB agmatine hydrolysis), and 3 (SpeC ornithine decarboxylation) should all be marked **covered**.

The one genuinely uncertain element is the **alternative two-step agmatine-deiminase branch** of the arginine route. KT2440 encodes agmatine deiminase AguA (PP_0266, K10536, EC 3.5.3.12), which converts agmatine to N-carbamoylputrescine, but the genome appears to lack an annotated downstream N-carbamoylputrescine amidohydrolase (**aguB / NCPAH, K12251, EC 3.5.1.53**) and has **no aguBA operon** analogous to the well-characterized *P. aeruginosa* system. This branch should be marked **candidate_uncertain (with an aguB gap)**: AguA alone

cannot complete the two-step route, and because the direct agmatinase SpeB already delivers putrescine from agmatine, the module remains satisfiable regardless of the deiminase branch status.

Two curation actions are important. First, **PP_4523**, labeled "Agmatinase" in local UniProt-derived metadata, is a **likely over-propagated annotation**: KEGG resolves it as guanidinobutyrase (K12255, EC 3.5.3.7), an arginine-catabolism ureohydrolase, not the biosynthetic agmatinase. It is a same-family paralog of the genuine agmatinase PP_2196 and should be demoted out of the module. Second, the KEGG `ppu00330` bucket is far broader than the putrescine module — roughly 33 of the 39 candidate genes belong to neighboring processes and must not be counted toward satisfiability. Critically, **no KT2440-strain-specific functional evidence** exists for any module gene; all calls rest on strong same-genus orthology transfer, primarily from *P. aeruginosa* PAO1 and *P. chlororaphis* O6.

2. Target-Organism Pathway Definition

2.1 Exact process included

The module covers **de novo putrescine (1,4-diaminobutane) supply from amino-acid precursors** in *P. putida* KT2440 via two alternative routes:

1. **Arg** → **agmatine** + **CO₂** — biosynthetic arginine decarboxylase (ADC / SpeA, EC 4.1.1.19).
2. **Agmatine** → **putrescine** — either:
3. **(2a) direct**: agmatinase (SpeB, EC 3.5.3.11): agmatine + H₂O → putrescine + urea; **or**
4. **(2b) two-step deiminase branch**: agmatine deiminase (AguA, EC 3.5.3.12) →
 N-carbamoylputrescine → *N*-carbamoylputrescine amidohydrolase (AguB, EC 3.5.1.53) →
 putrescine + CO₂ + NH₃.
5. **Ornithine** → **putrescine** + **CO₂** — ornithine decarboxylase (ODC / SpeC, EC 4.1.1.17).

2.2 Neighboring pathways to keep separate

The resolved KEGG bucket `ppu00330` ("Arginine and proline metabolism") is an aggregate map, not a module. The following co-bucketed processes are **outside** the module boundary and must not be counted toward putrescine satisfiability:

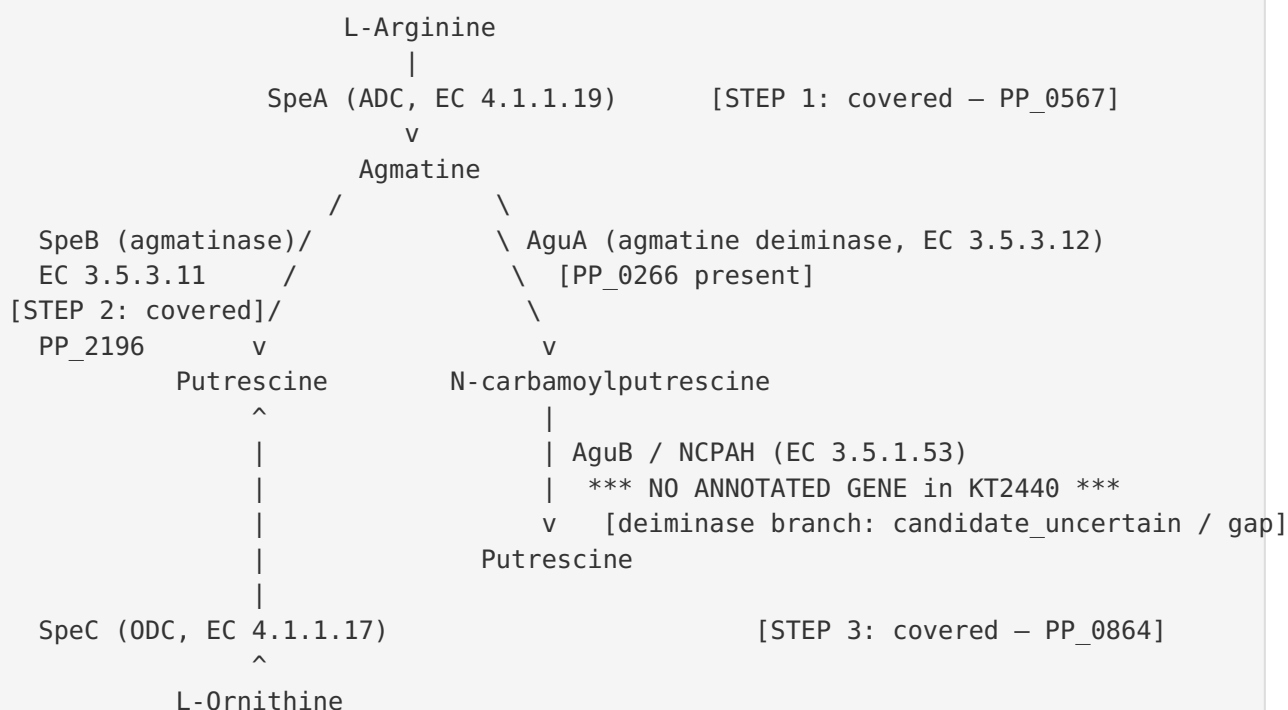
Adjacent process	Representative KT2440 candidate genes	Relationship to module
Arginine succinyltransferase (AST) catabolism	astA-I/PP_4479, astA-II/PP_4480, astB/PP_4477, astD/PP_4478, astE/PP_4475	Arginine → glutamate; not putrescine
Arginine oxidase / 2-ketoarginine route	aruH/PP_3721, aruI/PP_3723, kauB/PP_5278	Arginine catabolism → GABA; not putrescine
Proline biosynthesis / catabolism	proB/PP_0691, proA/PP_4811, proI/PP_5095, putA/PP_4947, pip/PP_5028	Proline metabolism (ppu00332/ppu00250)
5-Oxoprolinase	oplA/PP_3515, oplB/PP_3514	Glutathione cycle
Ornithine cyclodeaminase; acetylornithine AT	ocd/PP_3533, argD/PP_4481	Ornithine→proline / ornithine biosynthesis
Creatine/creatinine, opine oxidase	creA/PP_3667, ooxA/PP_4456, ooxB/PP_4457	Guanidino/opine catabolism
Downstream polyamine/ GABA aldehyde DHs	patD/PP_1481, prr/PP_2801, spuC-II/PP_5182, puuB/PP_2448, PP_2588, PP_2589	Putrescine/GABA catabolism — downstream of module
Other decarboxylases	ldcC/PP_4140 (Lys→cadaverine), nspC/PP_2929 (spermidine branch)	Different diamine/polyamine products
Guanidinobutyrase (mis-labeled "agmatinase")	PP_4523	Arginine catabolism ureohydrolase (see §5)

2.3 Alternate names and database definitions

- **SpeA** = biosynthetic arginine decarboxylase (ADC); EC 4.1.1.19; KEGG K01585. Distinguish from acid-inducible AdiA.
- **SpeB** = agmatinase (agmatine ureohydrolase); EC 3.5.3.11; KEGG K01480.
- **SpeC** = ornithine decarboxylase (ODC); EC 4.1.1.17; KEGG K01581.
- **AguA** = agmatine deiminase / iminohydrolase (AIH); EC 3.5.3.12; KEGG K10536.
- **AguB** = N-carbamoylputrescine amidohydrolase (NCPAH / CPA); EC 3.5.1.53; KEGG K12251. Note the frequent historical confusion with **ptcA** (putrescine transcarbamylase), a non-homologous, non-analogous enzyme found in Gram-positive bacteria.

- MetaCyc separates "putrescine biosynthesis I (ODC)," "II (agmatine → agmatinase)," "III (agmatine → AguA+AguB)," and "IV (arginine → ADC)." KEGG map = "Arginine and proline metabolism" ([ppu00330](#)).

3. Expected Step Model



Module step	Enzyme (EC)	KEGG KO	KT2440 gene	Call
1. Arg → agmatine	ADC / SpeA (4.1.1.19)	K01585	PP_0567	covered
2a. Agmatine → putrescine (direct)	Agmatinase / SpeB (3.5.3.11)	K01480	PP_2196	covered
2b-i. Agmatine → N-carbamoylputrescine	Agmatine deiminase / AguA (3.5.3.12)	K10536	PP_0266	present
2b-ii. N-carbamoylputrescine → putrescine	NCPAH / AguB (3.5.1.53)	K12251	none annotated (PP_3019?)	candidate_uncertain / gap
3. Ornithine → putrescine (direct)	ODC / SpeC (4.1.1.17)	K01581	PP_0864	covered

Net: Steps 1, 2 (via agmatinase), and 3 are **covered**. The optional deiminase branch (2b) is **incomplete** — treat as `candidate_uncertain`, not a module-blocking gap, because step 2 is already satisfied by agmatinase.

4. Candidate Genes and Evidence

4.1 High-confidence core module genes

SpeA — PP_0567 (Q88QC7), biosynthetic arginine decarboxylase, EC 4.1.1.19. KEGG orthology places PP_0567 in K01585, the biosynthetic ADC ortholog group; this is the committed first step of the arginine route, producing agmatine. UniProt annotates "Catalyzes the biosynthesis of agmatine from arginine" (evidence inferred from homology, PE3). Evidence type: **orthology, strong same-genus transfer**. The route it initiates is experimentally validated in *P. aeruginosa* PAO1, where putrescine is synthesized "indirectly from arginine via arginine decarboxylase (ADC; the speA product)" [PMID: 12634339], and in *P. chlororaphis* O6, *speA* contributes to the polyamine pool under GacS control [PMID: 28862813]. No KT2440-specific enzymatic assay exists. **Call: covered.**

SpeB — PP_2196 (Q88KU3), agmatinase, EC 3.5.3.11. KEGG assigns PP_2196 to K01480 (agmatinase, biosynthetic, agmatine → putrescine + urea) and it is the sole K01480 ortholog in ppu. This is the direct, single-enzyme completion of the arginine route and the reason the module is satisfiable independent of the deiminase branch. UniProt family: arginase family, agmatinase subfamily (IPR005925/IPR006035/IPR020855). Evidence type: **orthology, strong**. Curation caveat: PP_2196 shares identical InterPro signatures with the paralog PP_4523; KEGG correctly discriminates them (§5). **Call: covered.**

SpeC — PP_0864 (Q88PI6), ornithine decarboxylase, EC 4.1.1.17. KEGG assigns PP_0864 to K01581 (ODC), the sole K01581 ortholog; UniProt reaction "L-ornithine → putrescine + CO₂." This single enzyme constitutes the entire direct ornithine route. Local metadata lists a spurious primary bucket (`kegg:ppu04148`); the EC/KO are unambiguous and the bucket tag should be treated as an artifact. The route is experimentally established in *P. aeruginosa* PAO1: putrescine is synthesized "directly from ornithine by ornithine decarboxylase (ODC; the speC product)" and a *speA speC* double mutant is a putrescine auxotroph [PMID: 12634339]. Evidence type: **orthology, strong**. **Call: covered.**

AguA — PP_0266 (Q88R68), agmatine deiminase / iminohydrolase, EC 3.5.3.12. KEGG assigns PP_0266 to K10536; UniProt describes "hydrolysis of agmatine into N-carbamoylputrescine in the ADC pathway of putrescine biosynthesis." AguA is the first step of the alternative deiminase branch. In *P. aeruginosa*, the aguA product with the aguB product "biosynthetically convert agmatine to putrescine in the ADC pathway" [PMID: 12634339]. Evidence type: **orthology, strong for AguA function; weak for branch completeness in KT2440**. Presence of AguA does **not** imply the branch is complete — the downstream amidohydrolase is required and appears unannotated (§5). **Call: present, branch candidate_uncertain.**

4.2 Paralog / over-annotation cases

- **PP_4523 (Q88EE2)** — labeled "Agmatinase" (arginase family, agmatinase subfamily; same InterPro as PP_2196) but KEGG = **K12255 guanidinobutyrase (EC 3.5.3.7)**, an arginine-catabolism ureohydrolase acting on 4-guanidinobutyrate (→ GABA + urea). **Out of module.**
- **PP_3019 (Q88II0)** — "Carbon-nitrogen hydrolase family protein" (nitrilase superfamily; IPR003010, IPR050345); no KO. **Unverified candidate** for the missing NCPAH (AguB is a nitrilase-family enzyme; [PMID: 12634339]).

4.3 Summary table

Gene	Locus	UniProt	KO	Role in module	Evidence	Call
speA	PP_0567	Q88QC7	K01585	Arg→agmatine (step 1)	Orthology (strong)	covered
speB	PP_2196	Q88KU3	K01480	Agmatine→putrescine (step 2, direct)	Orthology (strong)	covered
speC	PP_0864	Q88PI6	K01581	Orn→putrescine (step 3)	Orthology (strong)	covered
aguA	PP_0266	Q88R68	K10536	Agmatine→N-carbamoylputrescine (branch)	Orthology (strong)	present branch uncertain
aguB (NCPAH)	—	—	K12251	N-carbamoylputrescine→putrescine	absent from annotation	gap
PP_4523	PP_4523	Q88EE2	K12255	mis-labeled "agmatinase"	KEGG = guanidinobutyrase	over-annotated → demoted
PP_3019	PP_3019	Q88II0	none	candidate NCPAH (nitrilase)	homology only, unverified	promoted to review

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 The deiminase branch is incomplete (a genuine gap)

In *P. aeruginosa*, the deiminase branch is a two-gene *aguBA* operon: AguA (agmatine deiminase) and AguB (N-carbamoylputrescine amidohydrolase), the latter "purified as a homohexamer of 33 kDa subunits" and structurally distinct from AguA [PMID: 12634339]. In KT2440:

- The genuine NCPAH ortholog group is **K12251 (EC 3.5.1.53)**. KEGG [link/ppu/K12251](https://www.kegg.jp/entry/K12251) and [link/ppu/ec:3.5.1.53](https://www.kegg.jp/entry/ec:3.5.1.53) return **no ppu gene**.
- The *aguA* locus PP_0266 has **no adjacent amidohydrolase** — its neighbors are unrelated (DctB kinase PP_0264, dTDP-rhamnose epimerase PP_0265, outer-membrane proteins PP_0267/0268). There is **no aguBA operon** as in *P. aeruginosa*.

- **Correction to Iteration 1:** an early assignment of *aguB* to PP_1639 was an **error** — PP_1639 (Q88MD4) is "Protein SprT" (KEGG K02742 = SprT), not NCPAH.
- PP_3019 (Q88II0), a carbon-nitrogen hydrolase / nitrilase-superfamily protein, is a **structurally plausible but experimentally unverified** candidate for the missing amidohydrolase.

Because agmatinase SpeB (PP_2196) already converts agmatine directly to putrescine, the module remains satisfiable. The deiminase branch is therefore recorded as **candidate_uncertain (with an aguB gap)**, not a module-breaking failure. Recent work shows this space is evolutionarily labile: non-homologous, convergently evolved NCPAHs ("AguY") in *Shewanella oneidensis* and *Microterricola gilva* are "as efficient as the canonical NCPAH from *Pseudomonas aeruginosa*" [PMID: 40673658], so a cryptic KT2440 amidohydrolase cannot be excluded without direct assay.

5.2 PP_4523 is a likely over-propagated "agmatinase"

UniProt annotates **both** PP_2196 (Q88KU3) and PP_4523 (Q88EE2) with the submission name "Agmatinase," the rule "Belongs to the arginase family. Agmatinase subfamily," and identical InterPro signatures (IPR005925, IPR006035, IPR020855). This is a classic paralog-driven over-propagation: family-level annotation cannot discriminate substrate specificity. KEGG splits them — **PP_2196 = K01480 agmatinase (EC 3.5.3.11)**, the true biosynthetic enzyme, versus **PP_4523 = K12255 guanidinobutyrase (EC 3.5.3.7)**. The distinction is further underscored by the literature record that *aguB/ptcA* and related ureohydrolase-family genes have "frequently been erroneously annotated in the past" [PMID: 21404211]. **Curation action: demote PP_4523 out of the putrescine module and re-label as guanidinobutyrase.**

5.3 Bucket-level over-inclusion

The `ppu00330` bucket contributed 39 candidates, but only ~5–6 are module-relevant. The remaining ~33 belong to arginine AST/arv catabolism, proline metabolism, 5-oxoprolinase, creatine/creatinine, opine oxidation, and downstream polyamine/GABA aldehyde dehydrogenases (§2.2). These are correctly annotated for their own pathways but must **not** be counted toward putrescine-module satisfiability — a boundary artifact of using a broad KEGG overview map as the starting bucket.

6. Mechanistic Model / Interpretation

KT2440 provides putrescine through a redundant, well-buffered supply system. Two independent precursors converge on the diamine:

- **Ornithine → putrescine** in a single decarboxylation (SpeC/PP_0864). This is the most direct route and, in the sister species PAO1, is biosynthetically sufficient by itself (a *speA speC* double mutant, not single mutants, is required to abolish putrescine — implying either route alone suffices) [PMID: 12634339].
- **Arginine → agmatine → putrescine**. SpeA (PP_0567) commits arginine, and agmatinase SpeB (PP_2196) completes the conversion in one step. A parallel deiminase branch (AguA/PP_0266 + a missing AguB) could in principle also complete it, but in KT2440 the amidohydrolase appears absent, so the arginine route is effectively single-track through agmatinase.

The functional consequence is that **the module is doubly redundant at the network level** (two precursors) but **single-track within the arginine branch** at the agmatine→putrescine node in KT2440, unlike PAO1 which additionally carries the operonic deiminase branch. This has a concrete curation implication: satisfiability should be scored on SpeA+SpeB+SpeC, and the deiminase branch should be flagged as a lineage-variable alternative that is *incomplete* here, so that presence of AguA alone does not auto-satisfy branch 2b.

The AguA present in KT2440 without an annotated AguB most plausibly functions in **agmatine catabolism/interconversion** rather than a complete biosynthetic branch — unless PP_3019 (or another nitrilase-family gene) supplies a cryptic NCPAH activity. Distinguishing these possibilities is the single most valuable experiment for this module.

7. Module and GO-Curation Recommendations

Step	Recommended status	Rationale
SpeA arginine decarboxylation	covered	PP_0567 = K01585, strong orthology
Agmatine → putrescine (agmatinase)	covered	PP_2196 = K01480, strong orthology
Agmatine → putrescine (AguA + AguB)	candidate_uncertain	PP_0266 present; AguB (K12251) not annotated; PP_3019 unverified
SpeC ornithine decarboxylation	covered	PP_0864 = K01581, strong orthology
PP_4523 "agmatinase"	remove from module	KEGG = guanidinobutyrase K12255
~33 neighboring-pathway candidates	not part of module	AST, proline, oxoprolinase, creatine, GABA catabolism

Module boundary verdict: The generic module boundaries are **correct** for KT2440. The module (SpeA→SpeB arginine route + SpeC ornithine route) is satisfiable, and treating the deiminase branch as an optional/alternative sub-branch is appropriate. **No module revision or new module document is required.** Add an optional note flagging the AguA deiminase branch as a lineage-variable alternative that is incomplete in KT2440 (aguB gap), to prevent auto-satisfaction from aguA alone. Do **not** add PP_1639 (SprT) or treat PP_4523 as an agmatinase step.

GO-curation notes: Existing terms suffice — GO:0033389 (putrescine biosynthesis from arginine), GO:0033388 (...from ornithine); activities GO:0008792 (ADC / PP_0567), GO:0008783 (agmatinase / PP_2196), GO:0047632 (agmatine deiminase / PP_0266), GO:0004586 (ODC / PP_0864), GO:0050126 (NCPAH, for any confirmed aguB). All assignments are ISS/orthology (no direct KT2440 evidence). No new GO term request is needed. Flag the PP_2196/PP_4523 pair for a paralog-split curation rule so "agmatinase" is not re-propagated to PP_4523.

8. Genes to Promote to Full fetch-gene Review

1. **PP_4523 (Q88EE2)** — HIGH. Resolve the agmatinase-vs-guanidinobutyrase conflict; confirm K12255/EC 3.5.3.7 and correct the UniProt-derived "Agmatinase" label; remove from module.
 2. **PP_3019 (Q88II0)** — HIGH. Test as the candidate N-carbamoylputrescine amidohydrolase (nitrilase family) that would complete the deiminase branch.
 3. **PP_0266 (aguA)** — HIGH. Resolve its physiological partner/role given the missing aguB; characterize genomic neighborhood and regulation.
 4. **PP_2196 (speB)** — MEDIUM. Confirm agmatinase vs arginase specificity (module step-2 enzyme).
 5. **PP_0864 (speC)** — MEDIUM. Confirm biosynthetic (vs biodegradative/inducible) ODC role in KT2440.
 6. **PP_0567 (speA)** — LOW/confirmatory. Core, high confidence.
-

9. Evidence Base and Key References

All module calls for KT2440 rest on **orthology transfer**, predominantly from *P. aeruginosa* PAO1; there is **no KT2440-strain-specific functional publication** for these genes. Species transfer within *Pseudomonas* is judged **strong** for core enzymatic assignments and **weaker** for pathway architecture details such as the presence/absence of an operonic deiminase branch.

PMID	Title (abbrev.)	Organism	How it supports/challenges the review
12634339	Identification of putrescine biosynthetic genes in <i>P. aeruginosa</i> ; characterization of agmatine deiminase and NCPAH	<i>P. aeruginosa</i> PAO1	Foundational. Defines both routes (ODC/SpeC, ADC/SpeA); AguA+AguB convert agmatine→putrescine; <i>speA speC</i> / <i>aguAB speC</i> mutants are putrescine auxotrophs; AguB is a nitrilase-family enzyme. Anchors orthology transfer to KT2440.
40673658	T6SS toxin Tse8 evolved from a novel NCPAH	<i>P. aeruginosa</i> , <i>Shewanella</i> , <i>Microterricola</i>	Documents three agmatine→putrescine routes and non-homologous, convergently evolved NCPAHs (AguY). Cautions that a cryptic KT2440 amidohydrolase cannot be excluded by homology alone.
21404211	Two enzyme families in putrescine synthesis from agmatine via agmatine deiminase	Multiple bacteria	Establishes that <i>aguB</i> (Gram-negatives) and <i>ptcA</i> (Gram-positives) are non-homologous and "frequently erroneously annotated" — directly relevant to the aguB gap and PP_4523 paralog caution.
28862813	Polyamine is a critical determinant of <i>P. chlororaphis</i> O6...	<i>P. chlororaphis</i> O6	<i>speA</i> and <i>speC</i> both contribute; <i>speAspeC</i> mutant eliminates polyamine production. Reinforces the both-routes model in a related pseudomonad.
20149107	Operon in agmatine metabolism regulating biofilm in <i>P. aeruginosa</i>	<i>P. aeruginosa</i> PA14	Shows agmatine-metabolism operon architecture (<i>aguBA</i> and <i>agu2ABCA'</i>) varies across strains — supports caution about assuming a KT2440 operon.
18721677	Comparative survey of putrescine from agmatine deamination	Multiple, incl. PAO1	Confirms agmatine→putrescine flux and regulation differ by organism.
17101165		Chloroviruses	Independent characterization of AIH (AguA-like) and CPA (AguB-like)

PMID	Title (abbrev.)	Organism	How it supports/challenges the review
	<i>Chlorella</i> viruses encode complete polyamine pathway		enzymes; corroborates two-step deiminase branch biochemistry.
31451546	Arginine biosynthesis modulates pyoverdine in <i>P. putida</i>	<i>P. putida</i>	Context for arginine flux in the target species; not directly on putrescine genes.
36922543	Polyamine-mediated oxidative-stress tolerance in <i>P. syringae</i>	<i>P. syringae</i>	Broader <i>Pseudomonas</i> context for polyamine physiology.

Database evidence (accessed 2026-08-13): KEGG orthology (rest.kegg.jp): ppu:PP_0567→K01585; PP_2196→K01480; PP_0864→K01581; PP_0266→K10536; PP_4523→K12255; K12251/EC 3.5.1.53 → **no ppu gene**; PP_1639→K02742 (SprT). UniProt (rest.uniprot.org): Q88KU3 & Q88EE2 = arginase family/agmatinase subfamily; Q88II0 = carbon-nitrogen hydrolase (nitrilase) family.

10. Limitations and Knowledge Gaps

- 1. No strain-specific functional data.** Every module call for KT2440 is inferred by orthology. Multiple targeted PubMed searches returned **zero** KT2440-specific functional papers on putrescine/polyamine biosynthesis, agmatinase, or the decarboxylases. The strongest experimental anchors are one genus away (*P. aeruginosa* PAO1; *P. chlororaphis* O6).
- 2. The aguB gap is unresolved.** Whether KT2440 has a functional deiminase branch depends on an amidohydrolase (K12251/EC 3.5.1.53) that is not annotated. PP_3019 is a plausible but untested candidate, and convergent AguY-type enzymes [PMID: 40673658] mean absence of a canonical homolog is not proof of absence of activity.
- 3. Paralog-driven over-annotation** (PP_2196 vs PP_4523) shows that UniProt family rules can propagate "agmatinase" to a guanidinobutyrase; the KEGG discrimination is trusted here but should be confirmed biochemically.
- 4. SpeC type** (biosynthetic vs acid-inducible biodegradative ODC) is not directly confirmed for KT2440.

5. **Bucket breadth.** The broad `ppu00330` map inflates the candidate list; module satisfiability must be judged on the ~5–6 relevant genes, not the 39.

11. Proposed Follow-up Experiments and Actions

- **Close the aguB gap (highest value):** Assay recombinant PP_3019 for N-carbamoylputrescine amidohydrolase activity; if negative, screen KT2440 nitrilase-superfamily and amidase-family loci. Alternatively, delete *speB* (PP_2196) and test residual agmatine→putrescine flux — residual activity would imply a functional deiminase branch.
 - **Genetic validation of core routes:** Construct KT2440 *speB* (PP_2196), *speC* (PP_0864), *speA* (PP_0567), and *speA speC* mutants; test for putrescine auxotrophy as in PAO1 to convert orthology calls into direct evidence.
 - **Confirm PP_4523 identity:** Enzymatic assay for guanidinobutyrase (EC 3.5.3.7) vs agmatinase (EC 3.5.3.11); update annotation and add a paralog-split curation rule.
 - **Confirm SpeC regulation/type:** Determine whether PP_0864 is constitutive/biosynthetic or acid-inducible under physiological conditions.
 - **Curation deliverables:** (a) mark steps 1–3 covered; (b) file aguB/NCPAH as a gap with PP_3019 flagged; (c) remove PP_4523 from the module; (d) exclude the ~33 neighboring-pathway genes from module scoring.
-

Uncertainty statement: Step calls rest on 1:1 KEGG orthology plus experimental validation in the sister species *P. aeruginosa* PAO1 and *P. chlororaphis* O6 (strong same-genus transfer). No KT2440-strain-specific functional study of *speA/speB/speC/aguA* (knockout, enzymology, or polyamine-pool measurement) was located — so all target-organism calls are homology-based inferences, not direct evidence. The completeness of the agmatine-deiminase branch and the identity of PP_3019 as NCPAH are open questions requiring biochemical confirmation.
